
PH102: Physics-2

Electrodynamics

Lecture # 06

Electrostatics-Electric potential and
electrostatic boundary conditions

25.01.2024

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Previous lecture

❖ Continuous charge distribution

The electric field of a line charge: $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\vec{r}')}{r^2} \hat{r} d\ell'$

The electric field of a surface charge: $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\vec{r}')}{r^2} \hat{r} da'$

The electric field of a volume charge: $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{r^2} \hat{r} d\tau'$

❖ Flux through any surface that encloses bunch of charges $= \frac{Q_{\text{enc}}}{\epsilon_0}$

❖ Gauss's law

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \quad (\text{Integral form})$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (\text{Differential form}) \quad \text{First Maxwell's Equation}$$

Electric Potential

The electric field $\mathbf{E}$ can be obtained

- (a) due to a charge distribution from Coulomb's law in general, or
- (b) by using Gauss's law, if the charge distribution is symmetric.

Another way of obtaining $\mathbf{E}$ is from electric scalar potential V .

To move a charge Q from a point $\mathbf{a}$ to $\mathbf{b}$ in the presence of an electric field $\mathbf{E}$, we need to apply some *external force*. The work done in displacing the charge by $d\mathbf{l}$ is

$$dW = \mathbf{F}_{ext} \cdot d\mathbf{l} = -\mathbf{F}_{Coul} \cdot d\mathbf{l} = -QE \cdot d\mathbf{l}$$

The minus sign implies the work is done by the external agent against the Coulomb force. **The total work done in moving Q from point A to B or the required potential energy is,**

$$W = -Q \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

Potential energy per unit charge (or, the potential difference between the points $\mathbf{a}$ and $\mathbf{b}$) is defined as

$$V_{ab} = V(\mathbf{b}) - V(\mathbf{a}) = \frac{W}{Q} = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

(Contd...)

$V_{ab} < 0 \Rightarrow$ a loss of potential energy and the work is done by the field.

$V_{ab} > 0 \Rightarrow$ a gain in potential energy and the work is done by the external agent.

If we take a standard point $A = \Theta$ at which the potential is known or agreed upon, then the potential at a position $B = r$ is

$$V(\mathbf{r}) = - \int_{\Theta}^r \mathbf{E} \cdot d\mathbf{l}$$

This is the electric potential at a position r .

Interestingly, the line integral is independent of the path and it depends only the end points. Thus, V_{ab} is path independent (**Why?**).

A special property of the electrostatic field

An electrostatic field $\mathbf{E}$ is a valid field if and only if

$$\nabla \times \mathbf{E} = 0.$$

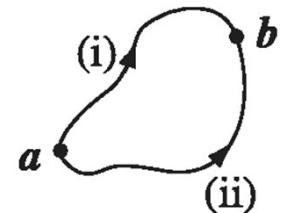
And hence it is a very special *kind of a vector*.

This special property of electric fields will reduce a *vector* problem (finding $\mathbf{E}$) to a much simpler *scalar* problem.

Applying Stroke's theorem to the curl of $\mathbf{E}$, the line integral of $\mathbf{E}$ over the closed loop is zero

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$

Therefore, along a specific path (say, along (i) or (ii)) we can write the expression of the form $V(\mathbf{r})$ or V_{ab} as in the earlier slides.



The potential difference between two points $\mathbf{a}$ and $\mathbf{b}$ is, therefore,

$$V(\mathbf{b}) - V(\mathbf{a}) = - \int_{\Theta}^b \mathbf{E} \cdot d\mathbf{l} - \left[- \int_{\Theta}^a \mathbf{E} \cdot d\mathbf{l} \right] = - \int_{\Theta}^b \mathbf{E} \cdot d\mathbf{l} - \int_a^{\Theta} \mathbf{E} \cdot d\mathbf{l} = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

According to the fundamental theorem of gradients, for a scalar field,

$$dV = \nabla V \cdot d\mathbf{l}$$

Therefore, the potential difference between the points **a** and **b** is

$$\int_a^b dV = V(\mathbf{b}) - V(\mathbf{a}) = \int_a^b \nabla V \cdot d\mathbf{l}$$

Comparing the two expressions for $V(\mathbf{b}) - V(\mathbf{a})$, we get (this is true for *any* points **a** and **b**, the integrands must be equal)

$$\int_a^b \nabla V \cdot d\mathbf{l} = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

$$\mathbf{E} = -\nabla V$$

If the line integral of $\mathbf{E}$ depended on the path taken, then the "definition" of V is meaningless. It simply would not define a function, since changing the path would alter the value of $V(\mathbf{r})$.

Comments on Potential

- Potential and Potential energy are completely different terms. Former is referred in the unit of Volt but the latter is measured in Joules.
- Potential is a scalar quantity and one can easily get $\mathbf{E}$ by taking gradient of V (vector problem is reduced to scalar problem without losing any information). $\mathbf{E}$ has three components and V has only one!!!

How can one function (V) possibly contain all the information that three independent functions (components of $\mathbf{E}$) carry?

The three components of $\mathbf{E}$ are not really independent.

They are explicitly interrelated by the very condition $\nabla \times \mathbf{E} = 0$, i. e.,

$$\frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}, \quad \frac{\partial E_z}{\partial y} = \frac{\partial E_y}{\partial z}, \quad \frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x}.$$

That is why $\mathbf{E}$ is a very *special* kind of a vector.

(Contd...)

- The choice of reference point is arbitrary and **does not affect the potential difference** between any two points as shown below. By changing the reference point to Θ' , the expression for potential $V'(\mathbf{r})$,

$$V'(r) = - \int_{\Theta'}^r \mathbf{E} \cdot d\mathbf{l}$$

It can be written in terms of the regular reference point Θ ,

$$V'(r) = - \int_{\Theta'}^r \mathbf{E} \cdot d\mathbf{l} = - \int_{\Theta'}^{\Theta} \mathbf{E} \cdot d\mathbf{l} - \int_{\Theta}^r \mathbf{E} \cdot d\mathbf{l} = K + V(r)$$

(K is a constant). Therefore, the potential difference remains unchanged.

$$V'(\mathbf{b}) - V'(\mathbf{a}) = V(\mathbf{b}) - V(\mathbf{a})$$

- The change in reference point **does not also affect the gradient** because the gradient of a constant is zero. $\nabla V' = \nabla V$.

(Contd...)

- The naturally chosen reference point in electrostatics is at $\Theta = \infty$, where the potential is 'set to zero', however, this convention fails when the charge distribution itself extends to infinity. The field of a uniformly charged plane is $(\sigma/2\epsilon_0)\hat{n}$. The potential at height z above the plane is

$$V(z) = - \int_{\infty}^z \frac{1}{2\epsilon_0} \sigma dz = -\frac{1}{2\epsilon_0} \sigma (z - \infty).$$

The remedy is simply to choose some other reference point.

- Potential obeys the superposition principle.

The force on test Q is $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots$

Dividing by Q , the electric field $\mathbf{E}$ is $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \dots$

Integrating from a common reference point the electric potential is

$$V = V_1 + V_2 + V_3 + \dots$$

- A surface over which the potential is constant is called an **equipotential**.

Poisson's equation and Laplace's equation

The electric field is, $\mathbf{E} = -\nabla V$.

What do the fundamental equations for E look like, in terms of V ?

From Gauss's law,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \nabla \cdot \mathbf{E} = -(\nabla \cdot \nabla V) = -\nabla^2 V = \frac{\rho}{\epsilon_0}$$

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{(Poisson's equation)}$$

In regions where there is no charge,

$$\nabla^2 V = 0 \quad \text{(Laplace's equation)}$$

Note that

$$\nabla \times \mathbf{E} = -(\nabla \times \nabla V) = 0.$$

$$\nabla \times \mathbf{E} = 0 \quad \text{permits} \quad \mathbf{E} = -\nabla V.$$

$$\text{in turn } \mathbf{E} = -\nabla V \quad \text{guarantees} \quad \nabla \times \mathbf{E} = 0.$$

Potential of a Localized Charge Distribution

Its easy to calculate V first and then calculate $\mathbf{E}$ by taking the gradient of V .

Poisson's relation relates V and ρ but it facilitates the determination of ρ from V .

However, it is ρ that we know and not V .

For a point charge at the origin,

$$\mathbf{E} = \left(\frac{q}{4\pi\epsilon_0 r^2} \right) \hat{r}$$

$$\mathbf{E} \cdot d\mathbf{l} = \left(\frac{q}{4\pi\epsilon_0 r^2} \right) \hat{r} \cdot dr\hat{r} = \left(\frac{q}{4\pi\epsilon_0 r^2} dr \right)$$

Setting the reference point at infinity, the potential due to a point charge q kept at the origin is

$$V(r) = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = \frac{-1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_{\infty}^r = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

The conventional minus sign in the definition of V was chosen precisely in order to make the potential of a positive charge comes out positive.

(Contd...)

In general, the potential of a point charge q is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Invoking the superposition principle, then, the potential of a collection of charges is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$

or, for a continuous distribution,

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r} dq.$$

And for various continuous distributions,

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{r} dl'$$

Line charge

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{r} da'$$

Surface charge

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r} d\tau'$$

Volume charge

Electrostatic boundary conditions

We have observed that the electric field (or for that matter, electric potential) inside and outside a region of charge distribution showed different values. This observation tells us that there could be a discontinuity at the boundary surface.

If the field exists in a region consisting of two different media, the conditions that the field must satisfy at the interface separating the media are called *boundary conditions*.

- These conditions are helpful in determining the field on one side of the boundary if the field on the other side is known.
- The conditions will be dictated by the types of material the media are made of.

We shall now look at the boundary conditions imposed on the electric field and the electric potential.

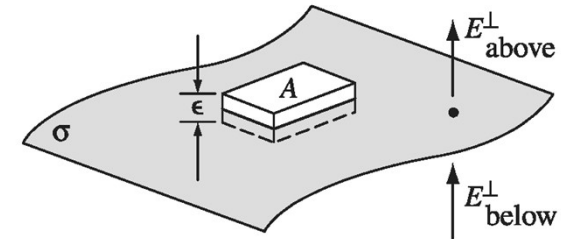
The normal component of E

The electric field is not continuous at a surface with charge density σ . Consider a Gaussian pill box of area A and thickness ϵ .

Gauss's law states that

$$\oint_s \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{enc} = \frac{1}{\epsilon_0} \sigma A$$

$$\oint_s \mathbf{E} \cdot d\mathbf{a} = \int \mathbf{E}_{above}^\perp \cdot \hat{n} da + \int \mathbf{E}_{below}^\perp \cdot (-\hat{n}) da = E_{above}^\perp A - E_{below}^\perp A$$



Therefore, $E_{above}^\perp - E_{below}^\perp = \frac{\sigma}{\epsilon_0}$

In the absence of surface charge $E^\perp$ is continuous.

$E^\perp$ denotes the component of E that is perpendicular to the surface immediately above or below.

The normal component of E is discontinuous by an amount σ/ϵ_0 at any boundary.

The sides of the pill box contribute nothing to the flux, in the limit as the thickness ϵ goes to zero.

The tangential component of $\mathbf{E}$

Consider a thin rectangular loop. The curl of the electric field states that

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$

$$\oint_P \mathbf{E} \cdot d\mathbf{l} = \int \mathbf{E}_{above}^{\parallel} \cdot \hat{l} dl + \int \mathbf{E}_{below}^{\parallel} \cdot (-\hat{l}) dl = E_{above}^{\parallel} l - E_{below}^{\parallel} l$$

Therefore, $E_{above}^{\parallel} = E_{below}^{\parallel}$ The ends contribute nothing as ϵ goes to zero.

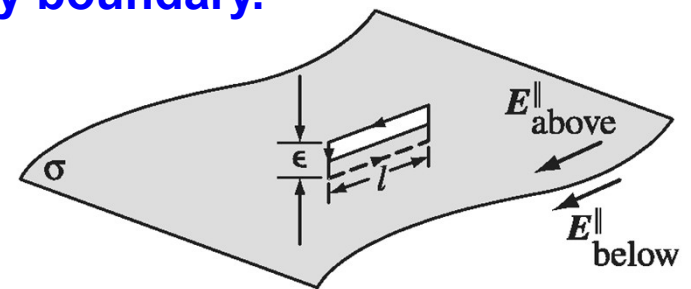
$E^{\parallel}$ denotes the component of $\mathbf{E}$ that is parallel to the surface immediately above or below.

The parallel component of $\mathbf{E}$ is continuous at any boundary.

The boundary condition of $\mathbf{E}$ can be combined in to a single formula

$$\mathbf{E}_{above} - \mathbf{E}_{below} = \frac{\sigma}{\epsilon_0} \hat{n}$$

$\hat{n}$ is the unit vector perpendicular to the plane pointing from 'below' to above.



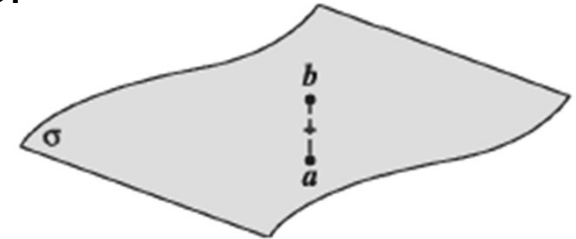
Boundary condition on electric potential

The potential across any boundary is continuous because, as shown below, as the path length shrinks to zero, the integral also goes to zero.

$$V_{above} - V_{below} = - \int_a^b \mathbf{E} \cdot d\mathbf{l} = 0$$

$$V_{above} = V_{below}$$

(Continuous)



However, the gradient of V inherits the discontinuity in $\mathbf{E}$,

$$\mathbf{E}_{above} - \mathbf{E}_{below} = -\nabla V_{above} + \nabla V_{below}$$

$$\nabla V_{above} - \nabla V_{below} = -\frac{\sigma}{\epsilon_0} \hat{n}$$

$$\left(\frac{\partial V_{above}}{\partial n} - \frac{\partial V_{below}}{\partial n} \right) = -\frac{\sigma}{\epsilon_0}$$

Please note that these boundary conditions relate the fields and potentials *just* above and *just* below the surface. The derivatives are the *limiting* values as we approach the surface from either side.

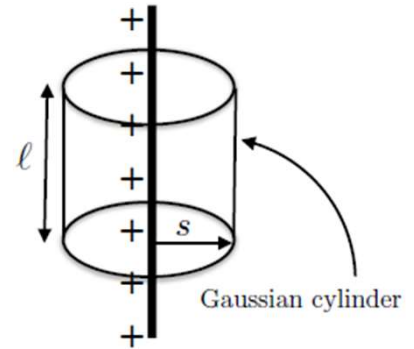
where $\frac{\partial V}{\partial n} = \nabla V \cdot \hat{n}$ denotes the **normal derivative** of V (that is, the rate of change in the direction perpendicular to the surface).

Solved example 1

Calculate electric potential due to an infinite line charge distribution λ .

Field can be evaluated by enclosing the line charge with a Gaussian cylinder of length ℓ and radius s .

The field only depends on distance from the line charge and directed away from it. Contribution to the flux from the top and bottom caps of the cylinder are zero.



$$\int \vec{E} \cdot d\vec{a} = |E| 2\pi s \ell = \frac{\lambda \ell}{\epsilon_0} \quad \text{or,} \quad \vec{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{s}$$

$$\vec{E} = -\vec{\nabla} V \implies V = -\frac{\lambda}{2\pi\epsilon_0} \ln s + \text{Constant}$$

fixed by reference point

Caution: Unlike the case of point charge, the reference point cannot be taken at infinity! Why?

Logarithm is undefined! Choose the reference point at $s = 1$ where $V = 0$.

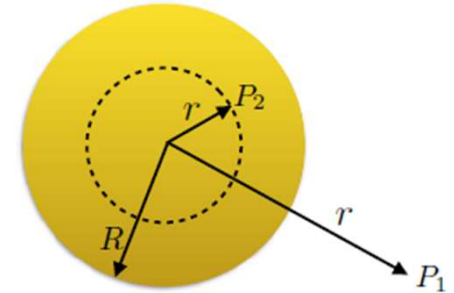
$$\therefore V = -\frac{\lambda}{2\pi\epsilon_0} \ln s \quad \text{with respect to reference point at } s = 1$$

Solved example 2

Potential due to uniformly charged sphere.

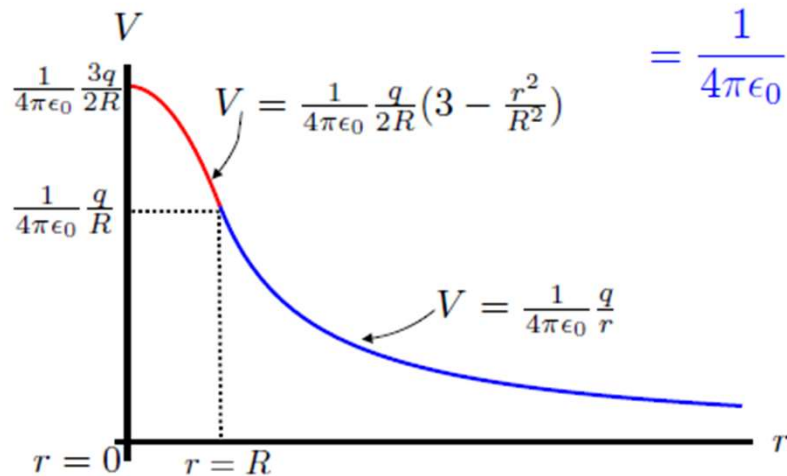
We know that

$$\vec{E} = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{qr}{R^3} \hat{r} & r < R \\ \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} & r > R \end{cases}$$



The electric potential at P_1 is $-\int_{\infty}^r \vec{E} \cdot d\vec{r}' = -\int_{\infty}^r \frac{q}{4\pi\epsilon_0 r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (r > R)$

The electric potential at P_2 is $-\int_{\infty}^R \vec{E} \cdot d\vec{r}' - \int_R^r \vec{E} \cdot d\vec{r}' = -\int_{\infty}^R \frac{q}{4\pi\epsilon_0 r'^2} dr' - \int_R^r \frac{qr'}{4\pi\epsilon_0 R^3} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{R} - \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} \frac{1}{2} (r^2 - R^2) = \frac{1}{4\pi\epsilon_0} \frac{q}{2R} \left(3 - \frac{r^2}{R^2} \right) \quad (r < R)$



Solved example 3

Potential due to uniformly charged spherical shell.

$$V(z) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma}{r} da \quad \text{where } r^2 = R^2 + z^2 - 2Rz \cos \theta'$$

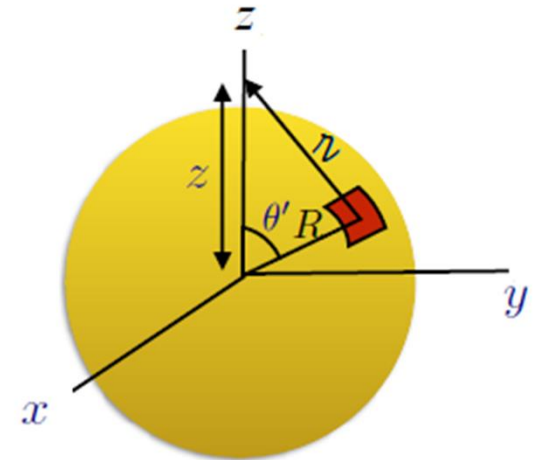
$$4\pi\epsilon_0 V(z) = \sigma \int_{\theta'=0}^{\pi} \int_{\phi'=0}^{2\pi} \frac{R^2 \sin \theta' d\theta' d\phi'}{\sqrt{R^2 + z^2 - 2Rz \cos \theta'}}$$

$$= 2\pi R^2 \sigma \int_{\theta'=0}^{\pi} \frac{\sin \theta' d\theta'}{\sqrt{R^2 + z^2 - 2Rz \cos \theta'}}$$

$$= 2\pi R^2 \sigma \left(\frac{1}{Rz} \sqrt{R^2 + z^2 - 2Rz \cos \theta'} \right) \Bigg|_0^{\pi}$$

$$= \frac{2\pi R\sigma}{z} \left(\sqrt{R^2 + z^2 + 2Rz} - \sqrt{R^2 + z^2 - 2Rz} \right)$$

$$= \frac{2\pi R\sigma}{z} \left(\sqrt{(R+z)^2} - \sqrt{(R-z)^2} \right)$$



In terms of r and the total charge on the shell, $q = 4\pi R^2 \sigma$,

$$V(r) = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{r} & (r \geq R), \\ \frac{1}{4\pi\epsilon_0} \frac{q}{R} & (r \leq R). \end{cases}$$

$$V(z) = \frac{R\sigma}{2\epsilon_0 z} [(R+z) - (z-R)] = \frac{R^2\sigma}{\epsilon_0 z}, \quad \text{outside}$$

$$V(z) = \frac{R\sigma}{2\epsilon_0 z} [(R+z) - (R-z)] = \frac{R\sigma}{\epsilon_0}, \quad \text{inside}$$

Note: outside the shell, $z > R$

Questions?